Problem Set 3, COSC330/PHYS306

Chapters 3 and 4

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Chapter 3

1. [Section 3-9, Exercise 39] Examine the conditions indicated in Figure 3–92, and identify the faulty gates.

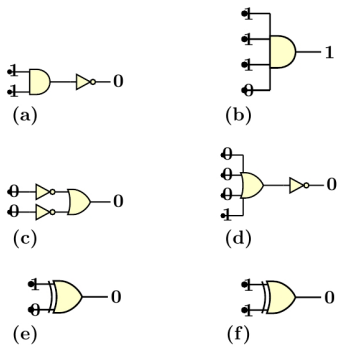


FIGURE 3–92

Answer : ***(b),(c) and(e) are faulty gates***

2. [Section 3-9, Exercise 40] Determine the faulty gates in Figure 3–93 by analyzing the timing diagrams.

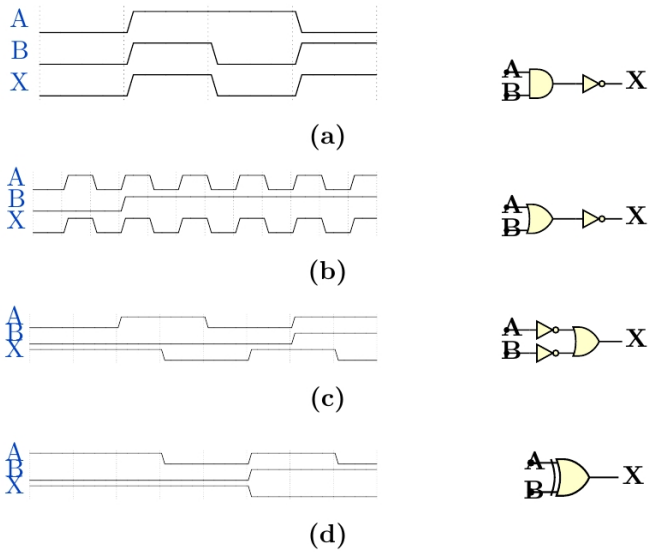


FIGURE 3–93

Answer: All of them

Chapter 4

3. [Section 4-3, Exercise 9] Apply DeMorgan's theorems to each expression:

(a) $\overline{A+B}$
(b) $\overline{\overline{A}B}$
(c) $\overline{A+B+C}$
(d) $\overline{ABC}$
(e) $\overline{A(B+C)}$
(f) $\overline{AB+CD}$
(g) $\overline{\overline{A}B+CD}$
(h) $\overline{(A+\overline{B})(\overline{C}+D)}$

A = $\overline{AB}$	E= $\overline{A}+\overline{BC}$
B= $A\overline{B}$	F =$(\overline{A+B})(\overline{C+D})$
C=$\overline{ABC}$	G = (A+B)(C+D)
D=$\overline{A}+\overline{B}+\overline{C}$	H = AB+CD

4. [Section 4-4, Exercise 13] Write the Boolean expression for each of the logic circuits in Figure 4–57.

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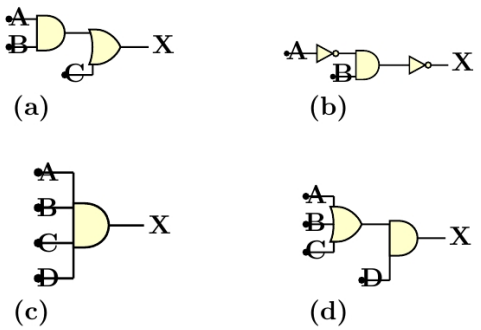
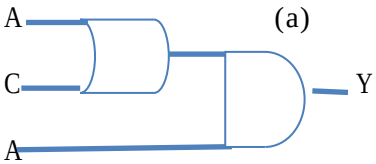


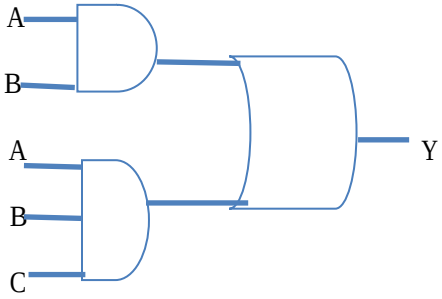
FIGURE 4–57

Answer: A = AB+C, $B=\overline{\overline{A}B}$ *C=ABCD* *D = (A+B+C)+D*

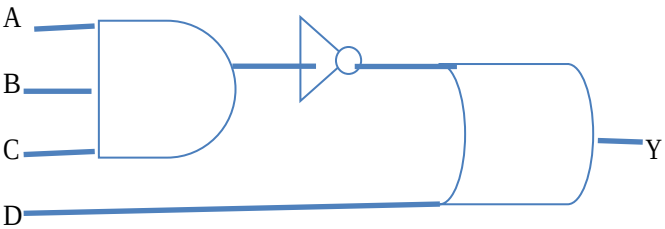
5. [Section 4-4, Exercise 15] Draw the logic circuit represented by each expression:

(a)

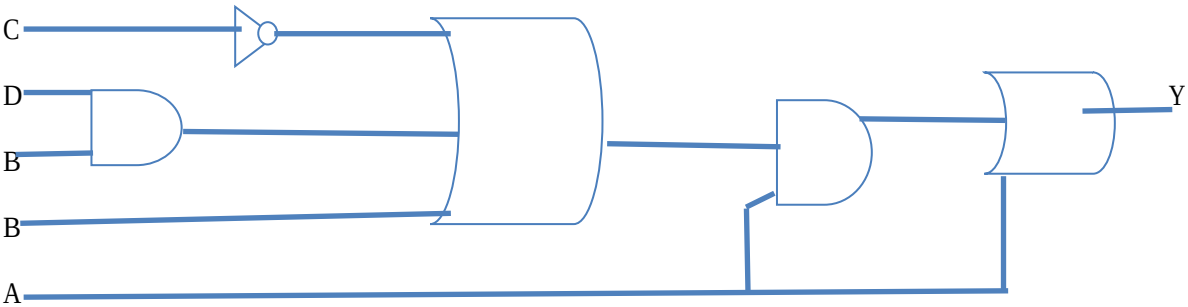




(C)



(D)



6. [Section 4-4, Exercise 18] Construct a truth table for each of the following Boolean expressions:

A	B	A+B
0	0	0
0	1	1
1	0	1
1	1	1

A	B	AB
0	0	0
0	1	0
1	0	0
1	1	1

(a) $A + B$

(b) AB

(c) $A + B(C + D)$

(d) $A + B + C$

(e) $AB + BC$

A	B	C	D	C+D	B(C+D)	A+B(C+D)
0	0	0	0	0	0	0
0	0	0	1	1	0	0
0	0	1	0	1	0	0
0	0	1	1	1	0	0
0	1	0	0	0	0	0
0	1	0	1	1	1	1
0	1	1	0	1	1	1
0	1	1	1	1	1	1
1	0	0	0	0	0	1
1	0	0	1	1	0	1
1	0	1	0	1	0	1
1	0	1	1	1	0	1
1	1	0	0	0	0	1
1	1	0	1	1	1	1
1	1	1	0	1	1	1
1	1	1	1	1	1	1

A	B	C	A+B+C
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

A	B	C	AB	BC	AB+BC
0	0	0	0	0	0
0	0	1	0	0	0
0	1	0	0	0	0
0	1	1	0	1	1
1	0	0	0	0	0
1	0	1	0	0	0
1	1	0	1	0	1
1	1	1	1	1	1

7. [Section 4-4, Exercise 22] Determine which of the logic circuits in Figure 4–59 are equivalent.

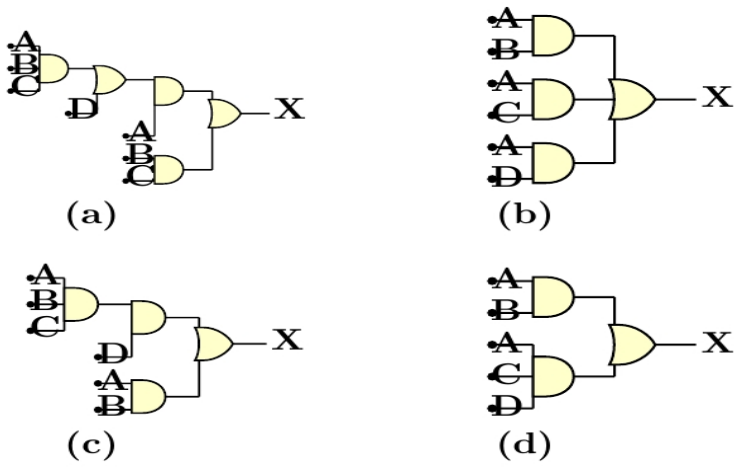


FIGURE 4–59

Answer: C and D are equivalent

8. [Section 4-7, Exercise 31] Develop a truth table for each of the following standard SOP expressions:

(a) $ABC + \overline{A}BC + AB\overline{C}$

(b) $XYZ + X\overline{Y}Z + \overline{X}YZ + \overline{X}Y\overline{Z} + X\overline{Y}\overline{Z}$

A	B	C	Output
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1

X	Y	Z	Output
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	1

9. **[Section 4-7, Exercise 36]** For each truth table in Table 4–15, derive a standard SOP and a standard POS expression.

ABC			X
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	1

(a)

ABC			X
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

(b)

$ABCD$				X
0	0	0	0	1
0	0	0	1	1
0	0	1	0	0
0	0	1	1	1
0	1	0	0	0
0	1	0	1	1
0	1	1	0	1
0	1	1	1	0
1	0	0	0	0
1	0	0	1	1
1	0	1	0	0
1	0	1	1	0
1	1	0	0	1
1	1	0	1	0
1	1	1	0	0
1	1	1	1	0

(c)

$ABCD$				X
0	0	0	0	0
0	0	0	1	0
0	0	1	0	1
0	0	1	1	0
0	1	0	0	1
0	1	0	1	1
0	1	1	0	0
0	1	1	1	1
1	0	0	0	0
1	0	0	1	0
1	0	1	0	0
1	0	1	1	1
1	1	0	0	1
1	1	0	1	0
1	1	1	0	0
1	1	1	1	1

(d)

A)

SOP (rows 001, 100, 101, 111):

X = A'B'C + AB'C' + AB'C + ABC

POS (rows 000, 010, 011, 110):

X = (A+B+C)(A+B'+C)(A+B'+C')(A'+B'+C)

B)

SOP (rows 101, 110, 111):

X = AB'C + ABC' + ABC

POS (rows 000, 001, 010, 011, 100):

X = (A+B+C)(A+B+C')(A+B'+C)(A+B'+C')(A'+B+C)

C)

SOP (rows where X=1: 0000, 0001, 0011, 0101, 0110, 1001, 1011, 1100):

X = A'B'C'D' + A'B'C'D + A'B'CD + A'BC'D + A'BCD' + AB'C'D + AB'CD + ABC'D'

POS (rows where X=0: 0010, 0100, 0111, 1000, 1010, 1101, 1110, 1111):

X = (A+B+C'+D)(A+B'+C+D)(A+B'+C'+D')(A'+B+C+D)(A'+B+C'+D)(A'+B'+C+D')(A'+B'+C'+D)(A'+B'+C'+D')

D)

SOP (rows where X=1: 0010, 0100, 0101, 0111, 1011, 1100, 1111):

X = A'B'CD' + A'BC'D' + A'BC'D + A'BCD + AB'CD + ABC'D' + ABCD

POS (rows where X=0: 0000, 0001, 0011, 0110, 1000, 1001, 1010, 1101, 1110):

X =
(A+B+C+D)(A+B+C+D')(A+B+C'+D')(A+B'+C'+D)(A'+B+C+D)(A'+B+C+D')(A'+B+C'+D)(A'+B'+C+D')(A'+B'+C'+D)

10. [Section 4-9, Exercise 40] Use a Karnaugh map to find the minimum SOP form for each expression:

(a) $\overline{A}\,\overline{B}C + A\overline{B}C + \overline{A}B\overline{C}$
(b) $AC(\overline{B} + C)$
(c) $\overline{A}(BC + B\overline{C}) + A(BC + B\overline{C})$
(d) $\overline{A}\,\overline{B}C + A\overline{B}\,\overline{C} + \overline{A}B\overline{C} + AB\overline{C}$

A) C'C C'C

A'B' 0 1 A'B 1 0

AB' 0 1 AB 0 0

Minimal SOP: B'C + A'BC'

B)

Multiply out: AC(B'+C) = AB'C + ACC = AB'C + AC

Minimal SOP: AC

C)

Factor: (A'+A)(BC+BC') = 1·B(C+C') = B·1

Minimal SOP: B

D)

C'C C'C

A'B' 0 1 A'B 1 0

AB' 1 0 AB 1 0

Minimal SOP: A'B'C + AC' + BC'